Lesson notes · for you

Malware: how it gets in

Cyber security · Level 3 · Lesson 1

Time
20 min

Materials
A phone (yours or a family phone, with permission); Paper and a pen for the checklist

Learning goals

- Name five kinds of malware and what each does.
- Explain the ways malware gets onto phones and computers.
- Choose the defence for each door, and know what to do if a device is infected.

Activity: Phone check-up

1. Warm-up (3 min): List every way an app or file has reached your phone or computer this month: the store, a chat group, a USB stick, a website, Bluetooth.
2. Doors (5 min): Mark each way as official (the store, the maker's own update) or risky (modded files, cracked games, shared sticks, pop-ups). Next to each risky one, write the defence that closes it.
3. Check-up (8 min): With permission, run the five-line check on a phone: automatic updates on, built-in protection on, "install unknown apps" off, no apps you don't recognise, backup on. Tick each line.
4. Share (4 min): Tell a friend or someone at home one thing you fixed, and why it matters.

Never remove apps from a shared family phone without asking. Never open files on a computer you suspect is infected.

Discussion

- Why do attackers make malware look like a gift?
- Why change passwords from a different device?
- Why is a backup the real cure for ransomware?

Check yourself

- One way malware gets in + the defence that closes it.

Try it this week
Give your phone a check-up: open Settings and check that automatic updates and built-in protection (Google Play Protect on Android) are on, that "install unknown apps" is off, and that there's no app you don't recognise. On a shared family phone, ask an adult before you remove anything.

Worksheet

Malware: how it gets in

Cyber security · Level 3 · Lesson 1

Name _____ Class _____ Date _____

1 **Your cousin offers you a cracked copy of a paid game on a USB stick. Do you install it?**

◯ No — cracked games often hide malware ◯ Yes — it's from family, so it's safe

2 **Match each kind of malware to what it does.**

Virus	• • • •	Pretends to be a useful app or hack
Worm	• • • •	Hides in a file and spreads when it's shared
Trojan	• • • •	Locks your files and demands payment
Ransomware	• • • •	Secretly records what you do
Spyware	• • • •	Spreads by itself across a network

3 **An app in the official store can never be malware.**

◯ True ◯ False

4 **Which of these could be signs that a phone has malware?**

☐ The battery runs down faster after a long video call ☐ Adverts pop up even on the home screen

☐ The data bundle runs out overnight while you sleep ☐ An app you don't remember installing

☐ The phone asks to install an update from Settings

5 **You think your phone has malware. Put what to do in order.**

☐ Tell your IT teacher or an adult at home

☐ Turn off Wi-Fi and mobile data

☐ Remove the app, or restore from a backup

☐ Change important passwords from a clean device

✎ **Which door is the most likely way malware could reach your phone or your family's computer? What one change will you make to close it?**

Check your answers

Malware: how it gets in

Cyber security · Level 3 · Lesson 1

1 **No — cracked games often hide malware**

Cracked software is one of the easiest ways in for malware.

2 **Virus — Hides in a file and spreads when it's shared**
Worm — Spreads by itself across a network
Trojan — Pretends to be a useful app or hack
Ransomware — Locks your files and demands payment
Spyware — Secretly records what you do

Knowing the kind tells you the defence: a worm is stopped by updates, a trojan by installing only from the official store, ransomware by a backup.

3 **False**

Official stores check apps and remove bad ones, which makes them far safer than random websites, but a few still slip through.

4 **Adverts pop up even on the home screen; An app you don't remember installing; The data bundle runs out overnight while you sleep**

Adverts on the home screen, an app you never installed and data used while you sleep all point to something running in secret.

5 **Top to bottom: 2, 1, 4, 3**

Cutting the connection first stops the malware spreading or sending your data.